Sporting Ethos — Solution Architecture

GigPoint Hackathon 2026 · Track 4 (Healthcare Tech) · Milestone 1 Project: Real-Time Patient Check-In, Live Queue, Ambient Charting & Pharmacy

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1. Business context

Sporting Ethos is a high-performance sports & healthcare centre specialising in sports science, sports medicine, rehabilitation, physiotherapy, nutrition, psychology, and athlete development. Patients book in advance and check in on arrival by scanning a QR code and submitting a few details.

2. Problem statement

Once a patient checks in, a confirmation email is generated as proof — but there is a **delay** before it appears. During that gap the healthcare expert has **no immediate, trustworthy signal** that the patient has arrived. Check-ins get overlooked → inaccurate consultation records, operational inefficiency, wasted appointment slots, and lost revenue.

Core problem: *the expert doesn't get fast, reliable confirmation that a patient has checked in — and that uncertainty costs the business.* The brief suggests real-time audio/visual confirmation and encourages going beyond it.

3. Solution overview

A single responsive web app (plus a small AI microservice) with **role-based views** that removes the uncertainty, turns the waiting room into a **live managed queue**, and extends into **ambient charting**, **pharmacy billing**, and **analytics** — the full clinic loop.

Confirmation is delivered on **four channels at once**, so it is impossible to miss:

1. **On-screen** — the patient's card flashes and appears at the top of the live queue.
2. **Audible** — a chime plays.
3. **Voice** — the browser speaks "*‹name› has arrived.*"
4. **Verifiable receipt** — a SHA-256-hashed digital proof for the patient.

Roles

Role	View	Purpose
Patient	`/checkin` (mobile, via QR)	Check in, get receipt + live position + "your turn", read visit summary
Reception	`/`	Live queue, add walk-ins, reports, QR, reception↔expert intercom
Expert / Doctor	`/expert` (code)	Current consultation, **ambient charting**, mark done, call reception
Pharmacy	`/pharmacy` (code)	Dispense by appointment ID, billing, inventory

4. System architecture

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PATIENT (phone)                          STAFF (laptop / tablet)
Universal QR → /checkin                  Reception /   ·  Expert /expert  ·  Pharmacy /pharmacy
• name, age, gender                      • live queue, metrics, voice+chime+highlight
• sequential APT-#### assigned           • status controls, emergency priority, add patient
• receipt + live position                • ambient charting, mark done, intercom
• "your turn" (voice, in language)       • pharmacy: match Rx → bill → payment → stock
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  • visit summary                          • reports: demographics, reasons, revenue, Excel
        │  INSERT / READ / REALTIME                │  READ / UPDATE / REALTIME / RPC
        ▼                                          ▼
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  │                        SUPABASE                                  │
  │  Postgres tables: checkins, medicines                            │
  │  Sequence: appt_seq  +  RPC next_appt_id()  (sequential IDs)     │
  │  Realtime: postgres_changes broadcast (<1s) + intercom broadcast│
  │  Row Level Security (kiosk-mode policies)                        │
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  AMBIENT CHARTING (separate microservice):
  dashboard audio ─► FastAPI (localhost:8000)
                       ├─ Groq Whisper-large-v3   → transcript
                       └─ Groq Llama 3.3 70B (JSON) → { summary, symptoms,
                                                        prescriptions, actions }
                    ─► saved to checkins.notes  +  consultation PDF  +  patient summary
```

Real-time: every screen subscribes to Supabase Realtime on `checkins`; an `INSERT` broadcasts and all dashboards re-render in **under a second** (no polling). Each subscriber uses a uniquely-named channel so multiple live views coexist. The reception↔expert **intercom** uses a separate Realtime *broadcast* channel (ephemeral, no DB writes).

5. Technology stack & rationale

Layer	Technology	Why
Frontend	React + Vite	Fast, component-based, instant HMR
Styling	Tailwind CSS	Clean, minimalist, light UI; shared `.card` / `.input` primitives
DB + realtime	Supabase (Postgres + Realtime)	Cross-device <1s updates with minimal backend code
Sequential IDs	Postgres sequence + `next_appt_id()` RPC	Guaranteed unique, ordered `APT-####`
Voice	Browser SpeechSynthesis (multi-lang)	Free, offline, zero dependency; always on
Receipt / bill	Web Crypto SHA-256 · jsPDF	Tamper-evident proof; branded PDF receipt & pharmacy bill
Ambient charting	FastAPI + **Groq** (Whisper + Llama 3.3 70B)	Free tier, blazing-fast transcription + structuring
Excel export	SheetJS (xlsx)	Real `.xlsx` for management reports
QR	qrcode.react	Renders + exports the universal check-in QR
i18n	Lightweight custom dictionary	English / Hindi / Telugu, no heavy dependency
Hosting (target)	Vercel (frontend) + Python host (backend)	One-click static deploy; public URL

6. Data model

`checkins` (one row per patient visit)

Column	Type	Purpose
`id`	uuid (pk)	Unique reference, on the receipt
`name`	text	Patient name
`appointment_id`	text	**Sequential** `APT-####` (auto-assigned)
`check_in_time`	timestamptz	Arrival time (queue order, latency, reports)

Column	Type	Purpose
`status`	text	`waiting` / `in_consult` / `done` / `left` / `no_show` / `paused`
`priority`	text	`normal` / `emergency` (emergency jumps the queue)
`gender`	text	Male/Female/Other — reports filter
`age`	int	Age → grouped Child/Teen/Adult/Senior
`source`	text	`self` (Pre-booked via QR) / `reception` (Walk-in, desk)
`hash`	text	SHA-256 of `appointment_id|name|check_in_time|id`
`notes`	jsonb	AI consultation notes {summary, symptoms, prescriptions, actions}
`pharmacy`	jsonb	Paid bill {bill_no, items[], total, paid, paid_at}

`medicines` (pharmacy inventory)

Column	Type	Purpose
`id`	uuid (pk)	Medicine ID
`name`	text	Medicine name
`price`	numeric	Unit price (₹)
`stock`	int	Units in stock (decremented on sale)
`category`	text	Category tag

Sequence `appt_seq` + **function** `next_appt_id()` → returns `APT-` + zero-padded next value.

Booking type (derived): `source = 'reception'` → **Walk-in**; else → **Pre-booked**.

7. Security & privacy

- Kiosk-style app; role screens (`/expert` , `/pharmacy`) are gated by simple access codes (env-configurable). For the hackathon, RLS allows anonymous read/write so any device works.
- The **Groq API key lives only on the FastAPI server** (`backend/.env`), never in the browser.
- No sensitive data in URLs; the universal QR carries only the check-in path.
- **Production hardening (roadmap):** phone-OTP patient auth, role-based RLS for reception/expert/pharmacy, per-clinic scoping, minimal PII retention, key rotation.

8. Scalability

- Cloud-native: Supabase scales DB + realtime; the frontend is static/CDN-served.
- Hardware: a phone (patient) + any screen (staff) — nothing to install.
- Multi-clinic: a `clinic_id` column + per-clinic realtime channel extends the same design from one room to a nationwide chain.

9. Measured impact (shown on-screen)

- **"Confirmed in <1 second"** — real scan→dashboard latency.
- Live counters — waiting, in-consult, done, LWBS, no-show.
- **Pharmacy revenue (₹)** and **top conditions** (from consultations) in Reports, exportable to Excel.